

Make Your PCB Design Ideas Fly High With **EAGLE**

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You set out to design your own printed circuit board (PCB) but are confused about which tool to use. How do you come to a decision? What are the factors you need to consider? In this article, we introduce you to EAGLE and tell you what you need to know to work with this tool.

EAGLE, a prototype of CadSoft Computers, is an easily applicable graphical layout editor. It is a flexible, expandable and scriptable electronic design automation application with

three modules, namely, schematic editor, layout editor and library editor, with identical user interfaces (UIs).

Designing your circuit has been made easy by access to design components in Farnell database, from within EAGLE environment via DesignLink, from within EAGLE environment, owing to the acquisition of CadSoft by Premier Farnell in 2009. Thus, you have everything you might need, in one place.

A new angle in simulation

EAGLE supports the newly-developed interface of PCB-Sim that lets you do the following:

1. Perform signal integrity analysis to ensure proper signal flow, even before proceeding to the layout stage
2. Calculate transmission line properties using the line calculator

3. Read S-parameter based data and convert to time-domain, with automatic correction of inaccuracies in the input data

4. Work between EAGLE based and SPICE based models with the addition of an integrated SPICE simulation engine. Also, see the circuit's functioning as waveforms

5. Easily import input-output buffer information specification models that describe the behaviour of an integrated circuit, thanks to SPICE engine

Route your design as EAGLE flies

Autorouting is fully integrated into the basic program. You can choose between manual or automatic routing option, and switch between the two easily. If you choose to go with the manual routing method, simply follow the follow-me-router tool that guides you at every stage. The smallest routing grid is 0.02mm in size. You can place components as you please, within the layout structure and use up to 16 signal layers. If you

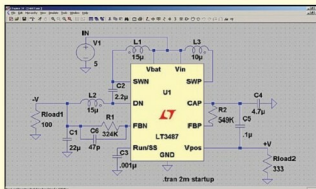


Fig. 1: Simulating with SPICE
(Image courtesy: chrisgammell.com/best-free-spice-program)

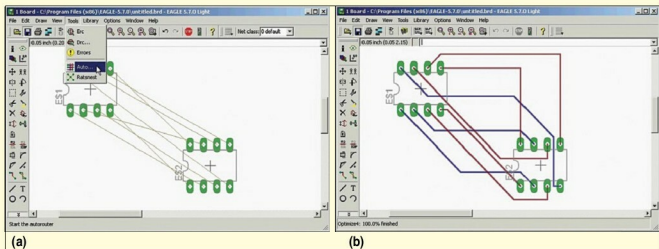


Fig. 2: Make your design route by itself: (a) before autorouting, and (b) after autorouting (Image courtesy: www.youtube.com)